PHYSICS MONSTER

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G.C.E A/L - 2024 MCQ ANSWER

UNOFFICIAL

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CORRECTED BY :

No of Correct responses

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01) (05) $\beta = 10\log\left(\frac{I}{I_0}\right)$

log $\rightarrow$ number.

02) $1MD = 10/20 = 0.5mm$

$L.C = 1MD - 1VD$

$= 0.5mm - \frac{19 \times 0.5}{20} = 0.025mm$

$L.C = \frac{0.5}{20} = 0.025mm$

(1)

03) கிடை வேகம் மாறிலி.

$\overrightarrow{u\cos\theta}$.

அதியுயர் புள்ளியில் கிடைவேகம் மட்டுமே காணப்படும்

$\frac{1}{2}mu^2\cos^2\theta = \frac{1}{4}\left(\frac{1}{2}mu^2\right)$

$\cos\theta = \pm\frac{1}{2}$

$\theta = 60^\circ$ (5)

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04) Newton's 3rd Law

(A) ✓

(B) ✗ திணிவுகளுக்கிடையிலான கவர்ச்சி விசை

(C) ✗ வெவ்வேறு பொருள்

Answer — 1

05) A) ✓ HP ஆனது தாக்கவல்லை

B) ✓ பொறிமுறை சக்தி இழப்படையும்

C) ✓ So, பொறிமுறை சக்தி காப்படையாது

But சக்தி காப்படையும்

Answer — 5

06) $\mu \quad e + r \longrightarrow \mu$

A) ✓

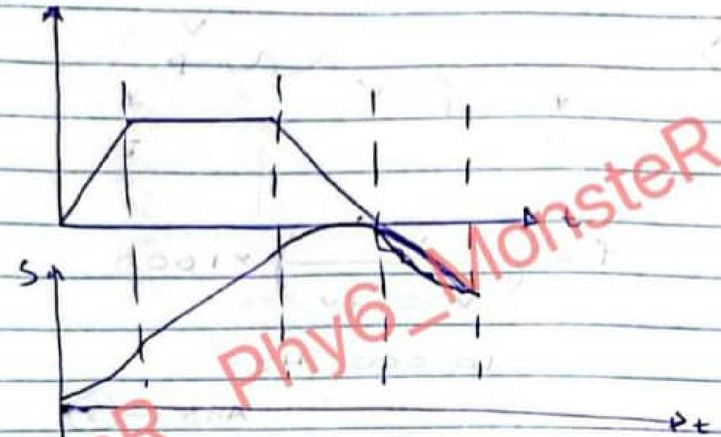
B) ✗ No quarks

C) ✓ $207 m_e$

3

No:
Date:

07)



Ans — 3

08)

$$\omega = \omega_0 + \alpha t$$

$$20 = 40 + \alpha(10)$$

$$\alpha = -2 \text{ rad s}^{-1}$$

$$\tau = -8 \times 2 = -16 \text{ Nm.}$$

Ans — 2

09) இயல்பான செப்பஞ் செய்கை.

அம்பம் முறைமை $\quad d = f_o + f_e$

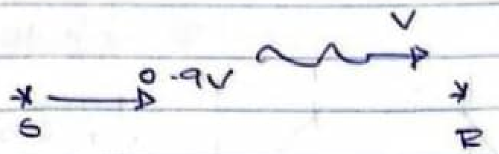
$M = \frac{f_o}{f_e} \Rightarrow 20 = \frac{80}{f_e} \quad = 84 \text{ cm.}$

$f_e = 4 \text{ cm.}$

Ans — 3

No:
Date:

10)



$$f' = \left(\frac{V}{V - 0.9V}\right) \times 1000$$

$= 10\,000$ Hz.

Ans — (5)

11) $E = \left(\frac{d\phi}{dt}\right)$

சக்தி காப்பு தத்துவம்

Example! –

Electrical Energy ⟶ mechanical

Motor

Reverse! Generator.

Ans — (2)

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πr^2

12)

$x = \frac{r}{3}$

(B)

Ans – (4)

13)

$I = \frac{P}{4\pi r^2}$ } புள்ளி முதல்

$I_2 > I_1$

$\Delta\beta = 10\log\left(\frac{I_2}{I_1}\right)$

$20 = 10\log\left(\frac{I'}{I}\right)$

Ans (5)

$I' = 100\,I$

14) Ideal ⇒ No power loss.

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\frac{110}{V_s} = \frac{200}{400}$$

$V_s = 220V$ (rms)

peak ⇒ $220\sqrt{2}$ V.

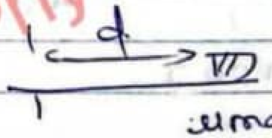
$V_p \cdot I_p = V_s I_s$

$I_{s\,rms} = \frac{110 \times 10}{220} = 5A$ (rms)

$I_{peak} = 5\sqrt{2}$

$(220, 5)$

Ans – 4

15) 

$\mu = 0.36$

$\mu mg = m d \omega^2$

$0.36 \times 10 = d \times \left(\frac{30 \times 2 \times 3}{60 \cdot 2}\right)^2$

$d = 3.6/9 = 0.4$ m

40cm Ans – 4

16) 

$f = 0.5 \times 100 = 50N$ $f = \mu(50)$

$F = ma$.

$100 - 40 - 50 = 10a$.

$a = 1\,ms^{-2}$

$F = ma$.

$40 - 50\mu = 5 \times 1$

$50\mu = 35$

$\mu = 7/10$

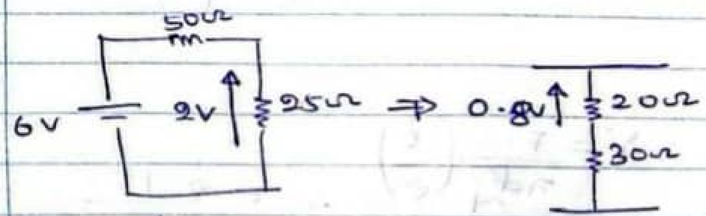
$= 0.7$

Ans – 1.

17) $\Delta l = l_0 \alpha \Delta\theta$

$5 \times 10^{-5} = 1 \times 10^{-5} \times \Delta\theta$.

$\Delta\theta = 5^\circ C$ — Ans 5

18) 

∴ R_f ⇒ 0.8 V.

Ans — 2

19) $v = \sqrt{\frac{T}{m}}$ රේඛීය ඝනත්වය m

වේ.

$v = f\lambda$ f = නියතයි.

$\lambda \propto \sqrt{T}$

$2 \propto \sqrt{20}$
$\lambda' \propto \sqrt{45}$

$\frac{\lambda'}{2} = \sqrt{\frac{45}{20}} = \sqrt{\frac{9}{4}}$

$\lambda' = 3\text{cm}$

Ans — (4)

20) $Y = \frac{F}{\pi d^2/4} \cdot \left(\frac{l}{e}\right)$ $\therefore e \propto \frac{1}{d^2 Y}$

$e = \frac{4Fl}{\pi d^2 Y}$

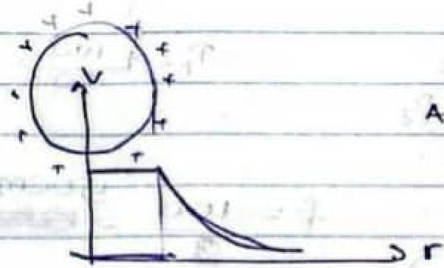
$e_A \propto \frac{1}{1 \times 2}$ $e_B \propto \frac{1}{4 \times 2}$

$e_C \propto \frac{1}{1 \times 1}$ $e_D \propto \frac{1}{1 \times 4}$

$\therefore e_C > e_A > e_D > e_B$

Ans — (3)

21)

22) 

Ans — (2)

23) $\downarrow \Delta P = \frac{4T}{R\uparrow}$

$P\updownarrow \rightarrow r\uparrow$

A) ~~U~~ $W = 2T \cdot A$

B) V $4\pi r^2 \rightarrow 16\pi r^2$

C) X $\frac{4}{3}\pi r^3 \rightarrow \frac{32}{3}\pi r^3$

8 மடங்கு.

Ans — 2

24) $P = I^2R$.

Ans — 3

25) $P = P_1 + P_2$

குவி
$P_1 = +\frac{100}{20} = +5D$

$+3D = +5D + P_2$

$P_2 = -2D$.

குழி

$f = \frac{100}{2} = 50cm$

Ans — 2

26) $10mV \rightarrow 40cm$

$\therefore 100cm \rightarrow \frac{10}{4} \times 10 \Rightarrow 25mV$

$= 0.025V$

R குறுக்கே $= 2 - 0.025$

$= 1.975V$

$\therefore 1.975 = \frac{R}{R+10} \times 2$

$19.75 = 0.025R$

$R = \frac{19750}{25}$

$= 790\Omega$ //.

Ans — 1

27) ${}^{235}_{92}U \rightarrow {}^{231}_{91}Pa + {}^{4}_{2}\alpha + {}^{0}_{-1}\beta + \bar{\nu}_e$

+ சக்தி

$\left.\begin{array}{l} 01 - \alpha \\ 01 - e^- \end{array}\right]$

Ans — 1

28) $75m^3 \rightarrow 0.04kgm^{-3}$ $R.H = 75$

$m = 3kg$

திருப்பு ஈரப்பதன் $= \frac{3}{75} \times 100 = 4kg$.

$\therefore$ extra $= 1kg$

Ans — 3

No:
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29)

$W = 0$

$\frac{q^2}{4\pi a} + \frac{q\varphi}{4\pi a} + \frac{q\varphi}{4\pi a} = 0$

$q + 2\varphi = 0$ Ans — (2)

$\varphi = -q/2$

30) h ஈர்ப்பின் கீழ் இயக்கம்

$U - mg = ma$

$a = \frac{(1-\beta)g}{\beta}$

$\uparrow v^2 = u^2 + 2as$

$v^2 = 2\left[\frac{1}{\beta} - 1\right]Hg$ Ans — (5)

$v^2 = u^2 + 2as$ $h = \frac{v^2}{2g}$

$0 = v^2 - 2gh$ $h = \left(\frac{1}{\beta} - 1\right)H$

No:
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31) A B } 3ம் நிலையம்

$d/2$ d

$\frac{d\theta}{dt}$

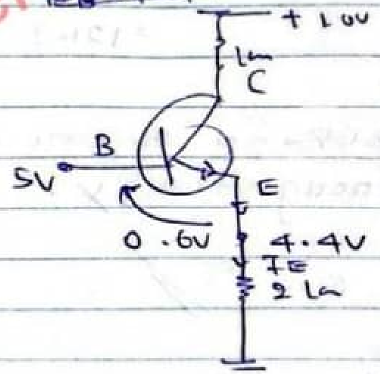
ms $R = kA(\theta - \theta_R)$ $R \propto \frac{1}{d}$

$\frac{4}{3}\pi r^3$ மாறிலி $4\pi r^2$ $R_A \propto 2/d$ $R_B \propto$

$R \propto A$

$R_A = 2R_B$

Ans — (4)

32) 

$I_E \simeq I_C$

$I_C = 2.2mA$

$V_{CE} = 10 - 3 \times 2.2$

$= 3.4V$

Ans — (2)

No
Date

33) $-10^\circ C \longrightarrow 0^\circ C \Big\} \; 100\times10^{-3}\times2\times10^{3}\times10$

$2kJ$

பனிக்கட்டி உருக $= 100\times10^{-3}\times3\times10^{5}$

$= 30kJ$.

நீர் ~~$30^\circ C$~~ $30^\circ C \longrightarrow 0^\circ C \Big\} = 100\times10^{-3}\times4\times10^{3}\times30$

$= 12kJ$.

∴ பனிக்கட்டி + நீர் கலவையே

BCZ, No enough energy.

∴ $\theta = 0^\circ C$

Ans — 2

34) (1) × இருக்கலாம், நிலை [illegible]

(2) × "

(3) ✓

(4) × BCZ [illegible]

(5) × உருகும்.

Ans — 3

No
Date

35) $\lambda = \dfrac{h}{P}$

மின் புலம்	காந்த புலம்
உந்தம் அதிகரிக்கும்	மாறாது.
ஆர்முடுகுவதால்	BCZ, வேகம் மாறாது (பருமன்)

∴ அலைநீளம் குறையும் ↓

∴ Ans — 3

36) $v = \dfrac{2a^2}{9\eta}(\rho-\sigma)g$

$8 \longrightarrow 2r$

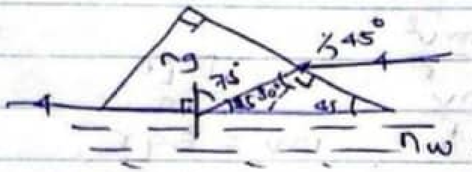
$8 \propto r^2$

$v \propto 4r^2$

∴ $v = 32\,cm\,s^{-1}$

Ans — 4

37)



$\sin45^\circ = n_g \sin30^\circ$

$n_g = \sqrt{2}$

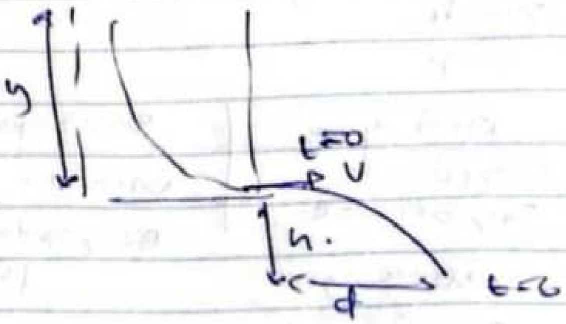
$\sin75^\circ = \dfrac{n_w}{n_g}$

$n_w = \sqrt{2}\sin75^\circ$

Ans — 2

No:
Date:

38)



$d = vt$

$h = \frac{1}{2} g t^2$

$t = \sqrt{\frac{2h}{g}}$

$v = d\sqrt{\frac{g}{2h}}$

$v = \sqrt{2gy} = d\sqrt{\frac{g}{2h}}$

$2gy = \frac{d^2 g}{2h}$ Ans — (3)

$y = \frac{d^2}{4h}$

39) $Bvq = \frac{mv^2}{r}$

$Bq = \frac{mv}{r}$

$v = \frac{Bqr}{m}$

$mv = Bqr$

$m_1 v_1 = B(q)(r)$

$m_2 v_2 = B(2q)(2r)$

(5)

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40) $F = A\bar{B} + BC$

Ans — (1)

41) $2T\sin 30^\circ = mg + T\sin 30^\circ$

$T\sin 30^\circ = mg$

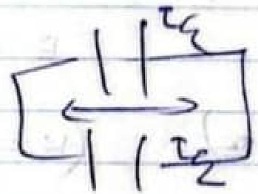
$3T\cos 30^\circ = mr\omega^2$

$\frac{3}{\tan 30} = \frac{r\omega^2}{g}$

$3\sqrt{3} = \frac{\sqrt{3}\,\omega^2}{g}$

$\omega = \sqrt{3g}$ Ans (3)

42)



A) ✓

B) ✗

C) ✗ (1/4)

Ans — (1)

No.
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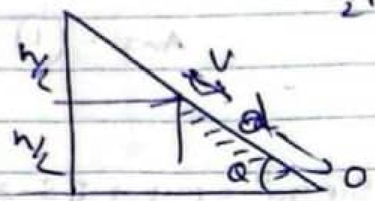
43)



$\frac{1}{2}mv^2 = mg\frac{h}{2}$

$\cancel{mg\frac{h}{2} = F \cdot d}$

$F \cdot d = mgh$

Ans — 3

$\mu mg\cos\theta \cdot d = mg\,2d\sin\theta$

$\boxed{\mu = 2\tan\theta}$

44) $K = \frac{Gm_1m_2}{2r}$ $V = -\frac{Gm_1m_2}{r}$

$E = \frac{-Gm_1m_2}{2r}$

$(2V = E)$ $(K = -E)$

Ans — 1

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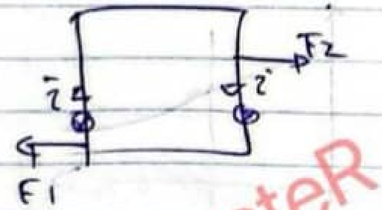
45)



$F_1 = \frac{\mu_0 I i b}{2\pi r}$ $F_2 = \frac{\mu_0 I i b}{2\pi(r+a)}$

$(F_1 - F_2) = \frac{\mu_0 I i}{2\pi}\left(\frac{b}{r} - \frac{b}{r+a}\right)$

$= \frac{\mu_0 I i ab}{2\pi r(r+a)}$

Ans — 4

46) $PV \propto T$

$6\times 3 \propto 600$

$4\times 9 \propto T$

$\frac{T}{600} = \frac{4\times 9}{6\times 3/2}$ Ans — 1

$T = 450\,K$

$\theta = 177^\circ C$

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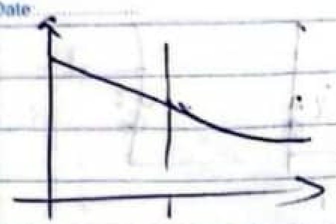
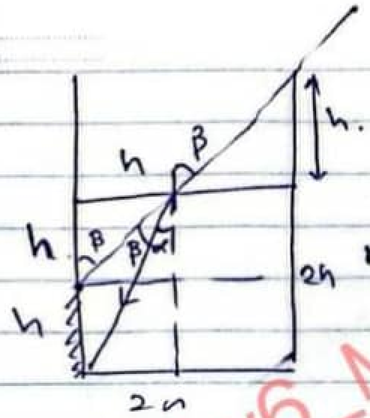
47)



Ans-(4)

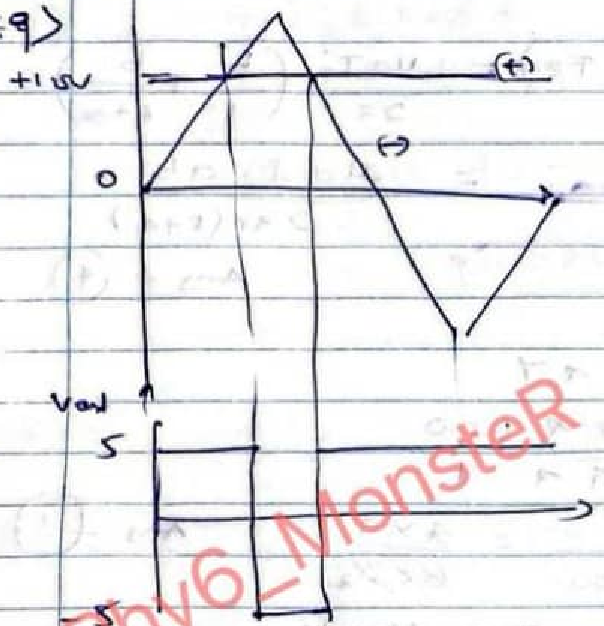
48)



$$n = \frac{\sin\beta}{\sin\alpha} = \frac{h/\sqrt{h^2+n^2}}{h/\sqrt{h^2+4n^2}} = \sqrt{5/2}$$

Ans-(2)

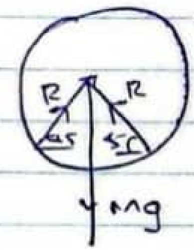
49)



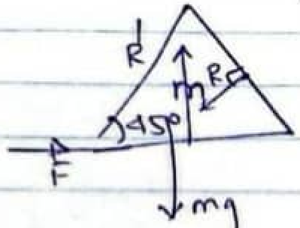
Ans (4)

$\Delta = 10^5$

50)



$2R\cos 45^\circ = mg$



$F = \mu(mg + R\cos 45)$

$F = R\sin 45^\circ$

$\frac{mg}{2} = \mu\left(mg + \frac{mg}{2}\right)$

(5)

$m = \frac{2m\mu}{1-\mu}$

21)

$(2 \times r) . T \times 2 = 0.04$

$$\boxed{\frac{1}{2r} = T}$$

வளை உள்ளேயும் வெளியும் உண்டு.

$\therefore$ Ans — (4).